Feature Extraction Techniques for Patterned Images: A Systematic Literature Review

Wenti Wahyuni
Dept. of Informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
email: wentiayu@students.amikom.ac.id
ORCID: 0009-0003-0330-3402

Arief Setyanto
Dept. of informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
email: arief_s@amikom.ac.id
ORCID: 0000-0003-0721-3941

Ema Utami
Dept. of informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
email: ema.u@amikom.ac.id
ORCID: 0000-0002-8237-8693

Kusrini Kusrini
Dept. of Informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
email: kusrini@amikom.ac.id
ORCID: 0000-0001-9573-3909

2024 2nd International Conference on Technology Innovation and Its Applications (ICTIIA) | 979-8-3503-5161-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICTIIA61827.2024.10761222

***Abstract*—This research aims to conduct a systematic review of feature extraction techniques in patterned images. A total of 47 journals were analyzed to identify the techniques used, their advantages and disadvantages, and the performance outcomes in various applications. The results indicate that statistical-based techniques such as the Gray Level Co-occurrence Matrix (GLCM), transformation-based techniques like discrete wavelet transform (DWT), visual descriptors such as SURF, and deep learning techniques like CNN are widely used and show excellent performance. For instance, statistical techniques are used for their simplicity and effectiveness. Additionally, deep learning techniques have succeeded in automatically analyzing image types from various fields based on neural network architectures, particularly CNN. Recent trends also show that combining traditional methods with deep learning, such as GLCM + Autoencoder and Haralick + Autoencoder, achieves the highest results, with a performance of 99.89%. This research states that although each technique has its pros and cons, deep learning techniques tend to be the primary choice for complex and high-precision feature extraction. The integration of classical methods with deep learning technology offers significant potential for further advancements in feature extraction for patterned images.**

Keywords: feature extraction, feature extraction techniques, image pattern

I. INTRODUCTION

In the current era, deep learning has achieved impressive performance in various visual recognition tasks [1], One crucial stage in image processing is feature extraction, which aims to filter important information from images that can be used for further analysis tasks such as object detection [2], [3], image classification [4], [5], or semantic segmentation [6]. Feature extraction is an essential part of data processing that provides a foundation for more complex tasks such as classification or clustering [7].

These techniques aim to determine the most relevant features in an image, thereby enhancing the performance of classification models. Features play a significant role in the classification process, as they can be utilized separately or combined with other features. The combination of features leads to an increase in the number of features, resulting in changes to the dataset size during the classification process [8]. Extracting features is useful in building datasets for pattern recognition [9].

Feature extraction in patterned images poses its own set of challenges due to the complex variations in scale, rotation, lighting, shape, and color. The variation of patterns in images, including differences in shape, size, and orientation, as well as the appearance of the same pattern at different scales and angles, is a common challenge. Additionally, variations in lighting can affect the appearance of patterns by altering contrast and color. Patterned images often contain noise, such as shadows, backgrounds, or other unwanted disturbances. This can make the feature extraction process more difficult for the algorithms used.

Therefore, effective feature extraction techniques must address these variations to produce relevant features. Although traditional techniques such as Principal Component Analysis (PCA) [10], [11] and Support Vector Machines (SVM) [12], [13], [14] have been widely used, recent developments in machine learning and deep learning offer more sophisticated and efficient approaches. Various feature extraction techniques have been developed in recent years, such as Scale-Invariant Feature Transform (SIFT) [15], Speeded-Up Robust Features (SURF) [16], [17], and Histogram of Oriented Gradients (HOG) [18]. The performance of these techniques can vary depending on the image data used and the characteristics of the images being processed.

With the multitude of techniques available, determining the most suitable method for a specific dataset or application can be quite challenging. Previous surveys on feature extraction methods have primarily focused on whole slide images (WSI) [19]. Therefore, the goal of this Systematic Literature Review (SLR) is to provide an overview of the feature extraction techniques used, identify the most effective methods, and highlight recent trends, advantages, disadvantages, and gaps in current research, particularly for patterned images. This SLR journal aims to serve as a reference for researchers and practitioners in developing better and more efficient feature extraction techniques for various applications. This SLR is expected to contribute to the advancement of knowledge in the field of image processing and analysis, as well as to encourage further innovation in the development of more sophisticated and adaptive feature extraction techniques.

979-8-3503-5161-3/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: Institut Teknologi Bandung. Downloaded on December 15,2024 at 05:40:28 UTC from IEEE Xplore. Restrictions apply.

II. Research Methodology

Systematic Literature Review (SLR) The purpose of this study is to gather and acquire knowledge from previous research. This discussion can serve as a foundation for understanding and developing future research in related fields [20]. To obtain knowledge from previous journals, specific steps were taken. The SLR consists of four stages [21]. These stages begin with identifying research problems, searching for journals, selecting the most relevant journals, and finally summarizing the key points from these journals. The stages can be seen in Figure 1.

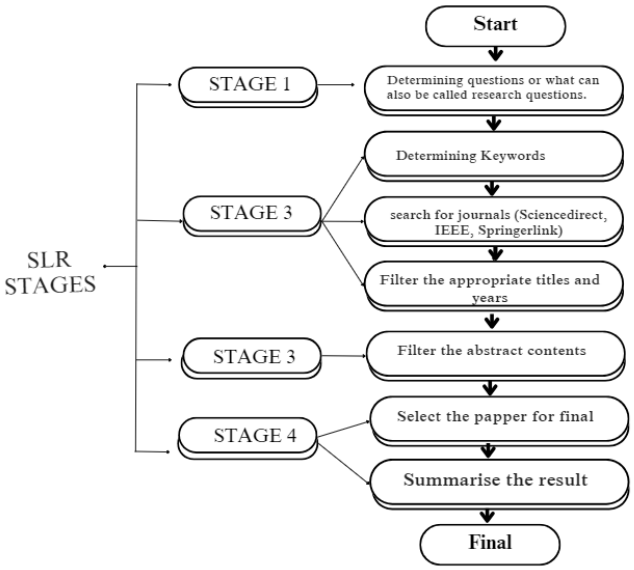


Fig. 1 Stages of SLR

Here are the steps undertaken:

A. Stage 1

Determine the research questions (RQs). In this study, four RQs were used. Initially, 10 RQs were designed, but after analysis, it was decided to select the four most relevant RQs for the study. These four RQs represent a summary of the scope of the original 10 RQs. The questions used in this study are:

1. What feature extraction techniques have been used in the analysis of patterned images?
2. What are the advantages and disadvantages of the feature extraction techniques identified in the literature?
3. How do these techniques perform in various applications?
4. What are the recent trends and future research directions in feature extraction for patterned images?

These questions were designed to determine whether the application of feature optimization techniques can improve classification results, identify commonly used methods, assess the performance of the various techniques, and understand which datasets are frequently used for feature optimization techniques.

B. Stage 2

At this stage, several steps were undertaken, including determining keywords and searching for journals. Keywords were used as search terms to find relevant journals. The databases used were:

1. ScienceDirect
2. IEEE Xplore
3. SpringerLink

The scope of the search was limited by filtering the years from 2021 to 2024 and using specific keywords. The keywords used were:

- "Feature Extraction AND Patterned Image Analysis"
- "Feature Extraction Techniques AND Image Classification"
- "Wavelet Transform AND Texture Analysis AND Images"
- "Deep Learning AND Image Feature Extraction"

Based on these keywords, a total of 442 journals were found. These were then filtered based on titles relevant to the research, resulting in 127 journals.

C. Stage 3

From these 127 journals, an analysis was then conducted based on their abstracts. This step involved reading the abstracts of the collected journals. If the abstract analysis revealed that the journal was not relevant, it was removed. After this selection process, 93 journals remained.

D. Stage 4

Following the abstract analysis, the contents of the remaining journals were thoroughly reviewed. Journals that did not meet the criteria were excluded. This step resulted in 47 journals. From these journals, key points were documented to answer the previously formulated research questions.

III. Results and Discussion

A. Description of Studies Reviewed

This study analyzes 47 scholarly journals published between 2021 and 2024. These articles were sourced from various academic databases such as ScienceDirect, IEEE Xplore, and SpringerLink. The reviewed studies include journal publications and conference proceedings, reflecting a significant increase in feature extraction research for patterned images over the past 4 years.

Table I. Distribution of Publication Years

Years	Number Of Studies
2021	3
2022	10
2023	15
2024	19

B. Feature Extraction Techniques

The review identifies several feature extraction techniques commonly used, such as GLCM, LBP, HOG, CNN, SURF, and various hybrid techniques combining different feature extraction methods. These techniques are categorized based on their approaches and methodologies, including statistical-based techniques, transformation-based techniques, machine learning-based techniques, visual descriptors, and color methods. Each category offers a unique approach to extracting relevant features from data, each with its strengths and weaknesses. Table II illustrates the categories of feature extraction techniques.

Authorized licensed use limited to: Institut Teknologi Bandung. Downloaded on December 15,2024 at 05:40:28 UTC from IEEE Xplore. Restrictions apply.

Table II. Classification of Feature Extraction Techniques

Types of Techniques	Feature Extraction Techniques	Description	Reference
Statistics-Based Techniques	PCA, LDA, LPP, GLCM, WLBP, SSF_HM, M2ECS-LBP and LBP	This technique uses statistical approaches to analyze and extract features from data. GLCM and LBP evaluate relationships between pixels, while other techniques project data into lower-dimensional spaces while preserving important information.	[14], [15], [18], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36]
Transformation Based Techniques	DWT, MI-WFE, IEWT, LSTP, DCT, TP-DWT, Wavelet Gabor and SECT	Transforming data from spatial domain to frequency domain for feature extraction. Existing techniques are highly effective in capturing details at various scales.	[12], [22], [37], [38], [39], [40], [41]
Deskriptor Visual	HOG, SIFT and SURF	This technique identifies and describes visual features from images. It is highly effective in detecting and describing features.	[15], [16], [17], [18], [30], [37], [42]
Color Method	HSV, RGB, LaB, and NIR	Utilizing color information in images for the feature extraction process.	[32], [43], [44]
Deep Learning-Based Techniques	CNN, ResNet, ResNet-18, ResNet-50, VGG, VGG-16, VGG-19, Incepton V3, Cross Vision Transformer, MEF-UNet, MTFE, GLCM + Autoencoder, Haralick + Autoencoder, FEE, and FRFENet, EfficientNetB3	This technique utilizes neural network architectures to automatically extract features from images.	[10], [33], [35], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60]

Based on the literature analyzed, the distribution of feature extraction technique usage indicates that statistical-based techniques are most frequently employed, with deep learning techniques, especially CNN, being the most popular (35%). This is followed closely by GLCM and LBP (33%). Visual descriptors such as SURF and HOG are utilized in 13% of cases, while transformation-based techniques, particularly DWT, are also popular (13%). Lastly, color methods are utilized in 6% of cases.

1. Statistics-Based Techniques

The techniques included in this group are widely used for texture analysis and dimensionality reduction of images. These techniques are applied across various types of images. Among them, GLCM and LBP are the most commonly used. These techniques provide a good description of local texture within images. This technique is applied due to its simplicity in processing, as it can describe image patterns with relatively straightforward calculations by analyzing the relationships between pixels. The image is converted to grayscale to facilitate the process and matching of pixel relationships. Despite its simplicity, the results obtained are quite accurate.

They have demonstrated high accuracy, nearly achieving close to 100% results in medical images [24] and 97% in textile images[14] using GLCM. Additionally, the performance of LBP feature extraction technique shows accuracy = 97.89%, F1-score = 97.87%, specificity = 99.87%, precision = 98.11%, and sensitivity = 97.94% [28] in soil textile. Another developed LBP technique, M2ECS-LPB, achieved impressive results of 99.98% in batik cloth, underscoring the significant impact of feature extraction algorithms on accuracy outcomes [34].

2. Transformation Based Techniques

The application of transformation techniques has shown promising results. For instance, the wavelet transform technique is used for feature extraction with repetitive patterns and can analyze images at multiple resolution levels. This is particularly useful for detecting features from different scale perspectives. Transformation techniques are also more resistant to noise and can reduce computational load. These techniques can be applied to various types of data, including images. Several studies that have applied transformation techniques for feature extraction include.

A comparison between statistical-based feature extraction techniques and transformation-based techniques indicates that LSTP performs better, particularly on data with complex textures [37]. Additionally, the use of transformation-based techniques has resulted in improved accuracy compared to previous methods, with an increase of 10.58%. Moreover, there has been an increase in precision by 6.52% and 14.23% [12]. Furthermore, the application of TP-DWT has also shown improvements [40]. Wavelet Gabor has demonstrated good results, achieving 96.15% accuracy, particularly in analyzing textile images [41].

3. Descriptor Visual

This technique is highly effective in capturing relatively stable local features and can detect and describe key points in an image, such as corners, edges, and specific textures. The technique focuses on identifying these important points and is very consistent with changes in scale and rotation. Matching and comparing features from other images are integral tasks of this technique. This is particularly useful for applications such as image retrieval and object detection.

The most widely used technique is HOG, which excels in extracting small objects [30] and edge contours [18]. Combining features based on HOG with other methods can enhance accuracy significantly; for instance, in [35], a combination with VGG-16 achieved 99.4% accuracy. Despite HOG's widespread application, the highest accuracy

Authorized licensed use limited to: Institut Teknologi Bandung. Downloaded on December 15,2024 at 05:40:28 UTC from IEEE Xplore. Restrictions apply.

results are obtained from the SURF technique, averaging above 99% and reaching a peak of 99.57% [16].

4. Color Method

In image analysis, color is a crucial aspect of object recognition applications. RGB, HSV, and Lab color models are particularly useful for analyzing color distributions in images. RGB typically shows lower performance compared to the other two methods. For example, [32], the HSV technique achieves higher results, reaching 77%. Applying the LaB method yields even higher results, at 95.4% and 97.4% [43]. However, when RGB is combined with other techniques, its performance can be significantly enhanced, achieving up to 99% accuracy when combined with NIR in textile image analysis [44].

5. Deep Learning-Based Techniques

These techniques offer powerful and flexible automatic feature extraction capabilities, particularly in pattern recognition. They utilize neural network architectures to automatically extract features from image data. These techniques include CNN, ResNet, VGG, and Inception V3. Based on the analysis, CNN is the most widely applied technique. Various fields have been analyzed using these techniques, such as Manufacturing [50], [56], [60], Cultural [53], [54], and others [46], [48]. Research using CNN has shown significant performance improvements. For instance, [46] achieved results of 93.693%, 94.716%, and 95.067%, demonstrating robustness against large-scale deformations and noise. In the same vein, a combination of GLCM, Autoencoder, and Haralick achieved 99.89% accuracy [33], while combining VGG-19 and ResNet-50 features processed with PCA and MVEC yielded the highest scores in accuracy, specificity, sensitivity, and precision: 98.51%, 97.58%, 99.49%, and 97.47%, respectively [10].

C. Recent Trends

Recent trends in feature extraction for patterned images show a shift towards the use of deep learning techniques and the integration of classical methods with deep learning. Techniques such as MEF-UNET [45] and Cross Vision Transformer [51] indicate a move towards more complex and adaptive neural networks. Integrating classical methods like GLCM and Haralick with Autoencoder shows potential for improving feature extraction accuracy and efficiency [33]. The use of techniques such as PCA, LDA, and LPP in combination with deep learning to reduce data dimensions without losing important information is also a focus of future research [27].

IV. Conclusion

This study has reviewed 47 research papers focusing on the techniques and methods used in feature extraction. Several commonly used techniques include statistical-based methods, transformations, visual descriptors, color methods, and deep learning-based techniques. Each technique has its strengths and weaknesses, depending on the image data and the analysis objectives. statistical methods (such as GLCM and LBP), transformation (DWT and DCT), visual descriptors (LBP and SURF), color methods, and deep learning techniques (CNN and other network architectures). CNN shows high accuracy up to 99.72%, SURF achieves 99.57%, and M2ECS-LBP obtains 99.89%. Combinations of techniques such as GLCM with DWT and DCT, HOG+LBP, as well as deep learning techniques like VGG-19 and ResNet 50 also show high results. Recent trends emphasize the integration of classical methods and deep learning to enhance accuracy and efficiency in patterned image analysis.

Overall, recent trends indicate an integration between classical methods and deep learning, alongside a growing focus on developing more efficient and adaptive neural networks. Future research is likely to continue exploring combinations of techniques to enhance feature extraction accuracy and efficiency, addressing challenges in patterned image analysis. In conclusion, while various techniques have their respective strengths and weaknesses, deep learning-based methods tend to be the preferred choice for applications requiring complex and high-precision image analysis. The integration of classical methods with deep learning technology holds significant potential for further advancements in feature extraction for patterned images.

References

[1] S. S. C. Z. L. X. W. L. W. C. Y. Z. Bao-Di Liu, "Few-shot image classification via hybrid representation," *Pattern Recognit*, vol. 155, 2024.

[2] S. Miao *et al.*, "Balanced single-shot object detection using cross-context attention-guided network," *Pattern Recognit*, vol. 122, 2022, doi: 10.1016/j.patcog.2021.108258.

[3] H. Chen *et al.*, "GasHis-Transformer: A multi-scale visual transformer approach for gastric histopathological image detection," *Pattern Recognit*, vol. 130, 2022, doi: 10.1016/j.patcog.2022.108827.

[4] Y. Zhou, X. Li, Y. Zhou, Y. Wang, Q. Hu, and W. Wang, "Deep collaborative multi-task network: A human decision process inspired model for hierarchical image classification," *Pattern Recognit*, vol. 124, 2022, doi: 10.1016/j.patcog.2021.108449.

[5] H. Chen *et al.*, "IL-MCAM: An interactive learning and multi-channel attention mechanism-based weakly supervised colorectal histopathology image classification approach," *Comput Biol Med*, vol. 143, 2022, doi: 10.1016/j.compbiomed.2022.105265.

[6] S. Yi, H. Ma, X. Wang, T. Hu, X. Li, and Y. Wang, "Weakly-supervised semantic segmentation with superpixel guided local and global consistency," *Pattern Recognit*, vol. 124, 2022, doi: 10.1016/j.patcog.2021.108504.

[7] M. Topolski and J. Kozal, "Novel feature extraction method for signal analysis based on independent component analysis and wavelet transform," *PLoS One*, vol. 16, no. 12 December, 2021, doi: 10.1371/journal.pone.0260764.

[8] A. A. Kasim, R. Wardoyo, and A. Harjoko, "The selection feature for batik motif classification with information gain value," in *Communications in Computer and Information Science*, 2017. doi: 10.1007/978-981-10-7242-0_9.

[9] M. Li, "Research on Textile Pattern Recognition Based on Artificial Intelligence," *2024 IEEE 7th Eurasian Conference on Educational Innovation (ECEI)*, pp. 335–338, 2024.

[10] M. M. Hossain *et al.*, "COVID-19 detection from chest CT images using optimized deep features and ensemble classification," *Systems and Soft Computing*, vol. 6, 2024, doi: 10.1016/j.sasc.2024.200077.

[11] O. P. Barus, J. Happy, Jusin, J. J. Pangaribuan, H. Samuel Zakaria, and F. Nadjar, "Liver Disease Prediction Using Support Vector Machine and Logistic Regression Model with Combination of PCA and SMOTE," in *Proceedings - 2022 1st International Conference on Technology Innovation and Its Applications, ICTIIA 2022*, 2022. doi: 10.1109/ICTIIA54654.2022.9935879.

[12] K. S. Rekha, D. Divya, M. J. Amali, and N. Yuvaraj, "Hybrid ML-MDKL feature subset selection and classification technique accompanied with rat swarm optimizer to classify the

Authorized licensed use limited to: Institut Teknologi Bandung. Downloaded on December 15,2024 at 05:40:28 UTC from IEEE Xplore. Restrictions apply.

multidimensional breast cancer mammogram image," *Optik (Stuttg)*, vol. 297, 2024, doi: 10.1016/j.ijleo.2023.171574.

[13] J. Lu *et al.*, "Lithology classification in semi-arid area combining multi-source remote sensing images using support vector machine optimized by improved particle swarm algorithm," *International Journal of Applied Earth Observation and Geoinformation*, vol. 119, 2023, doi: 10.1016/j.jag.2023.103318.

[14] T. Meeradevi and S. Sasikala, "Automatic fabric defect detection in textile images using a labview based multiclass classification approach," *Multimed Tools Appl*, 2024, doi: 10.1007/s11042-023-18087-7.

[15] C. E. Tezcan, B. Kiras, and G. Bilgin, "Classification of Histopathological Images by Spatial Feature Extraction and Morphological Methods," in *TIPTEKNO 2021 - Tip Teknolojileri Kongresi - 2021 Medical Technologies Congress*, 2021. doi: 10.1109/TIPTEKNO53239.2021.9632899.

[16] S. Katoch, V. Singh, and U. S. Tiwary, "Indian Sign Language recognition system using SURF with SVM and CNN," *Array*, vol. 14, 2022, doi: 10.1016/j.array.2022.100141.

[17] X. Liu, M. Chen, T. Xu, and H. Pengpeng, "Research on particle swarm optimized image alignment algorithm based on improved SURF features," in *Proceedings - 2022 Chinese Automation Congress, CAC 2022*, 2022. doi: 10.1109/CAC57257.2022.10055586.

[18] H. Yin, Y. Chen, J. Xiong, R. Xia, J. Xie, and K. Yang, "An improved local binary pattern method for pollen image classification and recognition," *Computers and Electrical Engineering*, vol. 90, 2021, doi: 10.1016/j.compeleceng.2021.106983.

[19] K. Al-Thelaya *et al.*, "Applications of discriminative and deep learning feature extraction methods for whole slide image analysis: A survey," *Journal of Pathology Informatics*, vol. 14. 2023. doi: 10.1016/j.jpi.2023.100335.

[20] Daniel, T. W. Cenggoro, and B. Pardamean, "A systematic literature review of machine learning application in COVID-19 medical image classification," in *Procedia Computer Science*, 2022. doi: 10.1016/j.procs.2022.12.192.

[21] Maniah, B. Soewito, F. Lumban Gaol, and E. Abdurachman, "A systematic literature Review: Risk analysis in cloud migration," *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 6. 2022. doi: 10.1016/j.jksuci.2021.01.008.

[22] S. Gudadhe, A. Thakare, and A. M. Anter, "A novel machine learning-based feature extraction method for classifying intracranial hemorrhage computed tomography images," *Healthcare Analytics*, vol. 3, 2023, doi: 10.1016/j.health.2023.100196.

[23] A. S. L. Erapaneni Gayatri, "Skin cancer classification enabled mobile neuro fuzzy network and entropy with weber local binary pattern based for feature extraction," *Expert Syst Appl*, vol. 252, 2024.

[24] T. P. P. K V Ranjitha, "Analysis on Improved Gaussian-Wiener filtering technique and GLCM based Feature Extraction for Breast Cancer Diagnosis," *Procedia Comput Sci*, vol. 235, pp. 285757–2866, 2024.

[25] D. L. Pinto *et al.*, "Image feature extraction via local binary patterns for marbling score classification in beef cattle using tree-based algorithms," *Livest Sci*, vol. 267, 2023, doi: 10.1016/j.livsci.2022.105152.

[26] B. M. Jebin, M. A. Rejula, and G. Eberlein, "Neonatal Seizure detection using GLCM feature extraction & AlexNet classification," *Multimed Tools Appl*, 2024, doi: 10.1007/s11042-024-18779-8.

[27] M. Imani, "Modified PCA, LDA and LPP feature extraction methods for PolSAR image classification," *Multimed Tools Appl*, vol. 83, no. 14, 2024, doi: 10.1007/s11042-023-17269-7.

[28] G. Sharmila and K. Rajamohan, "Soil classification using active contour model for efficient texture feature extraction," *International Journal of Information Technology (Singapore)*, vol. 15, no. 7, 2023, doi: 10.1007/s41870-023-01404-6.

[29] L. Li, J. Gao, H. Ge, Y. Zhang, and H. Zhang, "An effective feature extraction method via spectral-spatial filter discrimination analysis for hyperspectral image," *Multimed Tools Appl*, vol. 81, no. 28, 2022, doi: 10.1007/s11042-022-13121-6.

[30] H. , C. S. , S. H. et al. Lee, "Small Airborne Object Recognition with Image Processing for Feature Extraction," *International Journal of Aeronautical and Space Sciences*, 2024.

[31] K. C. Ahnaf and E. S. Wahyuni, "Comparative Analysis of Image Processing Methods using GLCM and LBP Feature Extraction for Lung Cancer Detection," *2023 6th International Seminar on Research of Information Technology and Intelligent Systems (ISRITI)*, pp. 513–518, 2023.

[32] A. D. Wibawa, E. Arif Wicaksono, S. D. Suryani, and R. Rumadi, "Javanese Batik Image Classification using Self-Organizing Map," in *ICCoSITE 2023 - International Conference on Computer Science, Information Technology and Engineering: Digital Transformation Strategy in Facing the VUCA and TUNA Era*, 2023. doi: 10.1109/ICCoSITE57641.2023.10127783.

[33] L. Li *et al.*, "Enhancing lung cancer detection through hybrid features and machine learning hyperparameters optimization techniques," *Heliyon*, vol. 10, no. 4, 2024, doi: 10.1016/j.heliyon.2024.e26192.

[34] A. H. Rangkuti, A. Harjoko, and A. Putra, "A Novel Reliable Approach for Image Batik Classification That Invariant with Scale and Rotation Using MU2ECS-LBP Algorithm," in *Procedia Computer Science*, 2021. doi: 10.1016/j.procs.2021.01.075.

[35] H. Abubakar, F. Al-Turjman, Z. S. Ameen, A. S. Mubarak, and C. Altrjman, "A hybridized feature extraction for COVID-19 multi-class classification on computed tomography images," *Heliyon*, vol. 10, no. 5, 2024, doi: 10.1016/j.heliyon.2024.e26939.

[36] U. S. Khwakhali, N. T. Tra, H. V. Tin, T. D. Khai, C. Q. Tin, and L. I. Hoe, "Fabric Defect Detection Using Gray Level Co-occurence Matrix and Local Binary Pattern," in *Proceedings - 2022 RIVF International Conference on Computing and Communication Technologies, RIVF 2022*, 2022. doi: 10.1109/RIVF55975.2022.10013920.

[37] H. B. F. D. C. S. K. S. P. R. A. Sinduja, "Detecting tumors in medical images using segmentation and feature extraction techniques," *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, vol. 9, 2024.

[38] K. R. D. Satyavati Jaga, "Brain tumor classification utilizing Triple Memristor Hopfield Neural Network optimized with Northern Goshawk Optimization for MRI image," *Biomed Signal Process Control*, vol. 95, 2024.

[39] G. H. K. D. S. C. S. G. Mohandass, "Lung Cancer Classification using Optimized Attention-based Convolutional Neural Network with DenseNet-201 Transfer Learning Model on CT image," *Biomed Signal Process Control*, vol. 95, 2024.

[40] N. Nagarani, R. Karthick, M. Sandra Carmel Sophia, and M. B. Binda, "Self-attention based progressive generative adversarial network optimized with momentum search optimization algorithm for classification of brain tumor on MRI image," *Biomed Signal Process Control*, vol. 88, 2024, doi: 10.1016/j.bspc.2023.105597.

[41] M. A. Rauf, M. Jehanzeb, U. Ullah, U. Ali, M. Kashif, and M. Abdullah, "Fabric Weave Pattern Recognition and Classification by Machine Learning," in *Proceedings - 2022 2nd International Conference of Smart Systems and Emerging Technologies, SMARTTECH 2022*, 2022. doi: 10.1109/SMARTTECH54121.2022.00026.

[42] M. R. M. A. A. Q. A. D. J. V. N. R. Puneet Thapar, "A hybrid Grasshopper optimization algorithm for skin lesion segmentation and melanoma classification using deep learning," *Healthcare Analytics*, vol. 5, 2024.

[43] D. B. P. I. R. D. F. A. D. Shofan Saifullah, "Palm Oil Maturity Classification Using K-Nearest Neighbors Based on RGB and L*a*b Color Extraction," *Procedia Comput Sci*, vol. 225, 2023.

[44] M. M. Khodier, S. M. Ahmed, and M. S. Sayed, "Complex Pattern Jacquard Fabrics Defect Detection Using Convolutional Neural Networks and Multispectral Imaging," *IEEE Access*, vol. 10, 2022, doi: 10.1109/ACCESS.2022.3144843.

[45] M. Xu, Q. Ma, H. Zhang, D. Kong, and T. Zeng, "MEF-UNet: An end-to-end ultrasound image segmentation algorithm based on multi-scale feature extraction and fusion," *Computerized Medical Imaging and Graphics*, vol. 114, 2024, doi: 10.1016/j.compmedimag.2024.102370.

[46] M. N. Vharkate and V. B. Musande, "Fusion Based Feature Extraction and Optimal Feature Selection in Remote Sensing Image Retrieval," *Multimed Tools Appl*, vol. 81, no. 22, 2022, doi: 10.1007/s11042-022-11997-y.

[47] U. Suttapakti, S. Srikamdee, and J. Onpans, "Multi-directional Texture Feature Extraction for Glaucoma Classification from Color Retinal Images," in *2022 19th International Joint Conference on Computer Science and Software Engineering, JCSSE 2022*, 2022. doi: 10.1109/JCSSE54890.2022.9836277.

Authorized licensed use limited to: Institut Teknologi Bandung. Downloaded on December 15,2024 at 05:40:28 UTC from IEEE Xplore. Restrictions apply.

[48] S. Madasamy, C. Nelson Kennedy Babu, V. Sumathi, and A. Chokkalingam, "A Robust Image Registration Using Multi-Scale Feature Extraction Techniques Using Deep Learning," in *Proceedings of the 4th International Conference on Smart Electronics and Communication, ICOSEC 2023*, 2023. doi: 10.1109/ICOSEC58147.2023.10276299.

[49] W. Zhu, D. Chen, Y. Quan, N. Li, and Y. Liu, "Full Range Feature Extraction Network for Hyperspectral Image Classification," in *Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing*, 2023. doi: 10.1109/WHISPERS61460.2023.10430853.

[50] D. Jia, "Intelligent Clothing Matching Based on Feature Analysis," in *Proceedings - 2022 14th International Conference on Measuring Technology and Mechatronics Automation, ICMTMA 2022*, 2022. doi: 10.1109/ICMTMA54903.2022.00135.

[51] A. F. Mohamed *et al.*, "Boosted Nutcracker optimizer and Chaos Game Optimization with Cross Vision Transformer for medical image classification," *Egyptian Informatics Journal*, vol. 26, Jun. 2024, doi: 10.1016/j.eij.2024.100457.

[52] N. Alruwais *et al.*, "Hybrid mutation moth flame optimization with deep learning-based smart fabric defect detection," *Computers and Electrical Engineering*, vol. 108, 2023, doi: 10.1016/j.compeleceng.2023.108706.

[53] S. Varshney, C. V. Lakshmi, and C. Patvardhan, "Madhubani Art Classification using transfer learning with deep feature fusion and decision fusion based techniques," *Eng Appl Artif Intell*, vol. 119, 2023, doi: 10.1016/j.engappai.2022.105734.

[54] T. Fan, H. Wang, and S. Deng, "Intangible cultural heritage image classification with multimodal attention and hierarchical fusion," *Expert Syst Appl*, vol. 231, 2023, doi: 10.1016/j.eswa.2023.120555.

[55] F. Atban, E. Ekinci, and Z. Garip, "Traditional machine learning algorithms for breast cancer image classification with optimized deep features," *Biomed Signal Process Control*, vol. 81, 2023, doi: 10.1016/j.bspc.2022.104534.

[56] A. Amelio *et al.*, "Defining a deep neural network ensemble for identifying fabric colors," *Appl Soft Comput*, vol. 130, 2022, doi: 10.1016/j.asoc.2022.109687.

[57] K. S. Kumar and M. R. Bai, "LSTM based texture classification and defect detection in a fabric," *Measurement: Sensors*, vol. 26, Apr. 2023, doi: 10.1016/j.measen.2022.100603.

[58] S. Zhao, R. Y. Zhong, J. Wang, C. Xu, and J. Zhang, "Unsupervised fabric defects detection based on spatial domain saliency and features clustering," *Comput Ind Eng*, vol. 185, 2023, doi: 10.1016/j.cie.2023.109681.

[59] N. Sajitha and S. P. Priya, "Automated fabric defect detection using hybrid particle cat swarm optimizer with deep learning model," *Multimed Tools Appl*, 2024, doi: 10.1007/s11042-024-18425-3.

[60] R. Liu, Z. Yu, Q. Fan, Q. Sun, and Z. Jiang, "The improved method in fabric image classification using convolutional neural network," *Multimed Tools Appl*, vol. 83, no. 3, 2024, doi: 10.1007/s11042-023-15573-w.

Authorized licensed use limited to: Institut Teknologi Bandung. Downloaded on December 15,2024 at 05:40:28 UTC from IEEE Xplore. Restrictions apply.